

WEIYI HE

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EDUCATION

Michigan State University

Ph.D. in Statistics, Dual Major in Computer Science

Aug. 2024 – Present

East Lansing, MI

University of Science and Technology of China (USTC)

B.S. in Statistics (Outstanding Graduate Award)

Sep. 2020 – Jul. 2024

Hefei, China

WORK EXPERIENCE

Amazon.com, Inc.

Applied Scientist Intern, AGI Foundations

Jun. 2026 – Aug. 2026

Arlington, VA

- Developed a novelty-search framework for generating diverse, answer-preserving input perturbations, enabling more discriminative robustness evaluation and revealing model fragilities missed by existing rewrite baselines.
- Characterized input- and weight-space robustness across 20+ large language models on mathematical reasoning benchmarks; found a strong cross-model association but divergent responses to post-training interventions, establishing the two axes as related yet non-interchangeable.

Articuler AI, Inc.

Generative AI Development Intern (Remote)

Jun. 2024 – Aug. 2024

San Francisco, CA

- Spearheaded the development of the **Search Agent** product — a scalable Python pipeline that automates professional background research for end users, from data collection to structured insight generation. (Deployed to production)
- Designed multi-step LLM workflows for profile parsing, query generation, information extraction, and validation using Google Search APIs and custom web scraping, improving retrieval quality and user experience.

Elven Technologies Pte. Ltd.

Data Engineering Intern (Remote)

Jan. 2024 – May. 2024

Singapore

- Built Python API integrations with major cryptocurrency exchanges (e.g., Coinbase, Binance, OKX) to automate transaction extraction and synchronization for Elven's Web3 financial reporting platform.
- Parsed and formatted transaction logs, enabling reliable downstream reporting and visualization in the platform.

RESEARCH AND OTHER PROJECTS

ReFT-based latent adversarial training on low-dimensional manifolds

Dec. 2025 – Apr. 2026

- Developed a defense strategy combining Low-rank Representation Fine-tuning (LoReFT) with Latent Adversarial Training (LAT); significantly mitigated Greedy Coordinate Gradient (GCG) attacks by implementing a multi-position intervention mechanism that prevents adversarial signal accumulation across causal sequences.
- Engineered a **Circuit-Aware** training framework utilizing Attribution Patching to isolate critical model circuits; optimized the inner-loop optimization by dynamically pruning non-essential gradient paths, achieving a 30% reduction in backward pass FLOPs while maintaining robust defensive performance.

Privacy Risks in LLM Agent Memory

Oct. 2024 – Feb. 2025

- Studied privacy risks in LLM agent memory and proposed **MEXTRA (Memory EXTRAction Attack)**, a black-box framework using targeted attacks and automated prompt generation under varying attacker knowledge.
- Evaluated memory vulnerabilities across real-world agent architectures and deployment settings; MEXTRA extracted at least one private item in **83%–90%** of targeted trials across two agents.

PUBLICATIONS & MANUSCRIPTS

- Weiyi He**, Yuping Lin, Jiliang Tang, Yue Xing. “Efficient LLM Adversarial Training via Low-Rank Defense and Circuit-Guided Surrogates.” *arXiv preprint arXiv:2607.28959*, 2026.
- Weiyi He**, Yue Xing. “Impact of Positional Encoding: Clean and Adversarial Rademacher Complexity for Transformers Under In-Context Regression.” *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026.
- Bo Wang, **Weiyi He**, Shenglai Zeng, Zhen Xiang, Yue Xing, Jiliang Tang, and Pengfei He. “Unveiling Privacy Risks in LLM Agent Memory.” *Proceedings of the Association for Computational Linguistics (ACL)*, 2025.

TECHNICAL SKILLS

Languages & ML: Python, R, C, PyTorch, TensorFlow, Hugging Face, scikit-learn, NumPy, Pandas

Tools & Research: Linux, Git, Slurm, AWS Bedrock, vLLM, TRL; LLMs, Trustworthy AI, Statistical Learning